

The UNIVERSITY OF AZAD JAMMU & KASHMIR

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LAB 04 : STUDENT CASE STUDY

UML Sequence Diagram: ATM Cash Withdrawal System

1. Background and Scenario

A commercial bank provides an Automated Teller Machine (ATM) that allows customers to withdraw cash from their bank accounts. A customer inserts an ATM card, enters a personal identification number (PIN), and requests a withdrawal. The ATM communicates with the bank system to authorize the transaction. The bank system checks the account and determines whether the requested amount can be withdrawn.

For this exercise, the customer wants to withdraw PKR 5,000. The ATM must not release cash unless the bank has validated the PIN and approved the withdrawal. When the transaction cannot be completed, the ATM displays an appropriate message instead of dispensing cash.

2. System Participants

Participant	Role in the Interaction
Customer	Initiates the transaction, provides the PIN, and requests a withdrawal.
ATM	Receives customer input, communicates with the bank, displays results, and dispenses cash after approval.
BankSystem	Validates the PIN, coordinates withdrawal authorization, and responds to the ATM.
Account	Provides balance-related information and supports debiting the account when permitted.

3. Functional Requirements

- FR-01: The customer inserts an ATM card and enters a PIN.
- FR-02: The ATM sends the card information and PIN to the BankSystem for validation.
- FR-03: Only after valid PIN confirmation may the customer request withdrawal of PKR 5,000.
- FR-04: The ATM forwards the withdrawal request to the BankSystem.
- FR-05: The BankSystem asks the Account whether sufficient funds are available.
- FR-06: If sufficient funds are available, the BankSystem requests the Account to debit PKR 5,000 and waits for confirmation.
- FR-07: After bank approval, the ATM dispenses PKR 5,000 and shows a success message.
- FR-08: If funds are insufficient, the BankSystem refuses the transaction and the ATM displays an insufficient-funds message; no debit or cash dispensing occurs.
- FR-09: If the PIN is invalid, the ATM informs the customer and does not continue to the withdrawal request.

4. UML Sequence Diagram

The following single sequence diagram shows the valid-PIN flow, the successful withdrawal, the insufficient-funds alternative, and invalid-PIN handling. Messages are read from top to bottom.

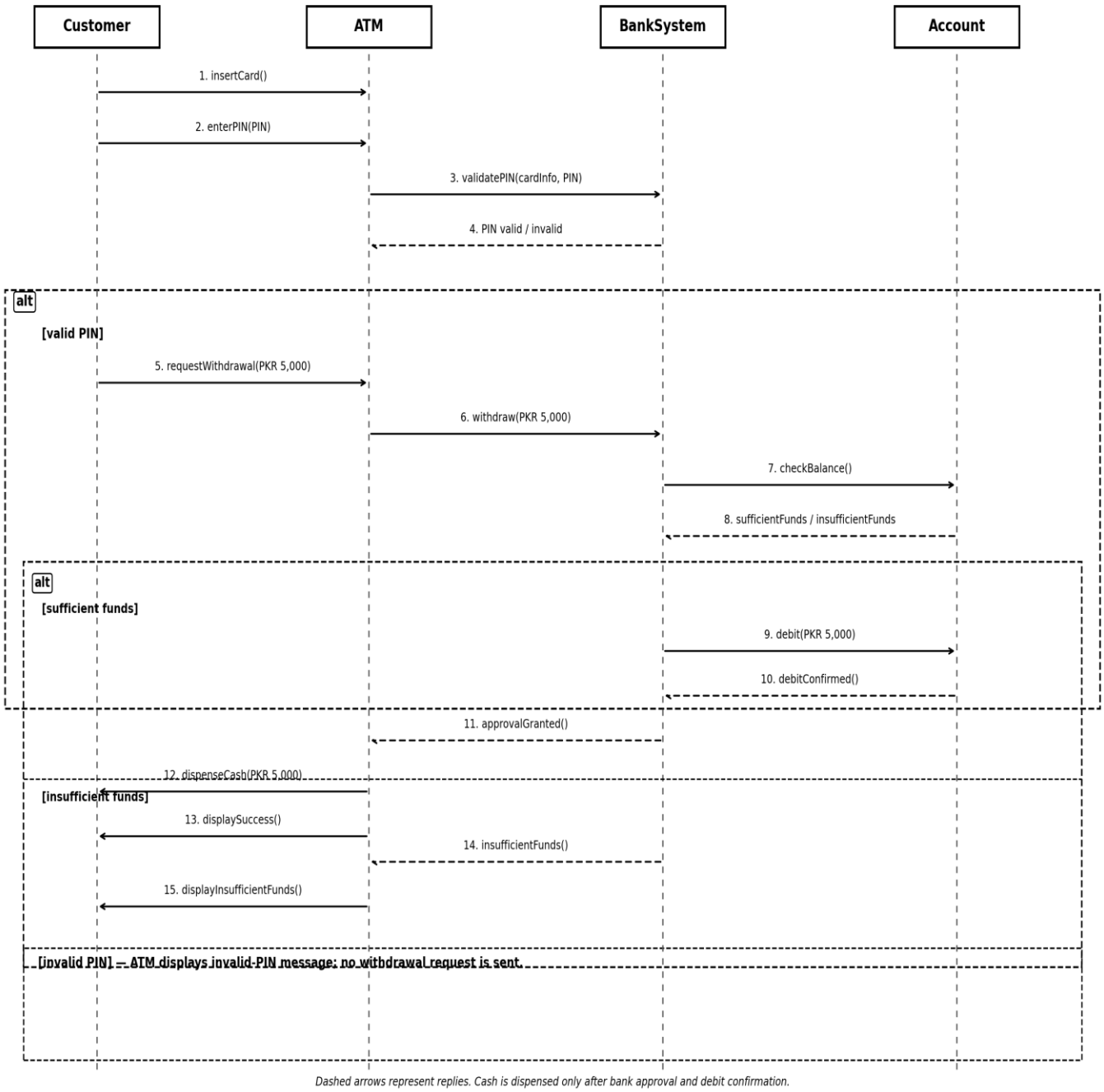


Figure 1. ATM cash withdrawal sequence diagram (PKR 5,000).

UML notation used: solid horizontal arrows represent requests/calls; dashed horizontal arrows represent replies. The alt combined fragment separates guarded alternative outcomes. The invalid-PIN guard stops the transaction before a withdrawal request.

Message Sequence (Chronological)

1. Customer → ATM: insertCard().
2. Customer → ATM: enterPIN(PIN).
3. ATM → BankSystem: validatePIN(cardInfo, PIN).
4. BankSystem ⇢ ATM: PIN valid / invalid.
5. [valid PIN] Customer → ATM: requestWithdrawal(PKR 5,000).
6. ATM → BankSystem: withdraw(PKR 5,000).
7. BankSystem → Account: checkBalance().
8. Account ⇢ BankSystem: sufficientFunds / insufficientFunds.
9. [sufficient funds] BankSystem → Account: debit(PKR 5,000).
10. Account ⇢ BankSystem: debitConfirmed().
11. BankSystem ⇢ ATM: approvalGranted().
12. ATM → Customer: dispenseCash(PKR 5,000), then displaySuccess().
13. [insufficient funds] BankSystem ⇢ ATM: insufficientFunds(); ATM → Customer: displayInsufficientFunds(). No debit or cash dispensing occurs.
14. [invalid PIN] ATM → Customer: displayInvalidPIN(); no withdrawal request is sent.

5. Answers to Questions to Guide the Design

Who initiates the first message?

Ans: The Customer initiates the interaction by inserting the ATM card.

Which participant is responsible for validating the PIN?

Ans: BankSystem validates the PIN after receiving the card information and PIN from the ATM.

Which participant knows whether enough money is available?

Ans: Account provides the balance-related information; it returns whether sufficient funds are available.

Which message must occur before the ATM can dispense cash?

Ans: The ATM must receive bank approval after the account has been debited and debit confirmation has been returned. Cash is dispensed only after approval.

What should the ATM do if the Account returns an insufficient-funds result?

Ans: The BankSystem refuses the transaction and the ATM displays an insufficient-funds message. There must be no debit and no cash dispensing.

Which part of the diagram would change if the customer entered the wrong PIN?

Ans: The PIN-validation reply would indicate invalid PIN. The ATM would display an invalid-PIN message, and the withdrawal request and all later withdrawal messages would not occur.

6. Modelling Constraints and Assumptions

- The withdrawal amount is fixed at PKR 5,000 throughout this case study.
- The ATM is assumed to have sufficient physical cash and the network is assumed to be available; no cash-stock or connectivity branches are modelled.
- The diagram focuses on message ordering. Activation bars and low-level implementation details are optional and are not required.
- No class attributes, database tables, or data-flow arrows are included because this is a sequence diagram.
- Assumption: after the BankSystem receives debit confirmation, it sends approval to the ATM; the ATM dispenses cash only after receiving that approval.
- Assumption: invalid-PIN handling is shown as an alternative outcome of PIN validation and terminates the withdrawal flow.

7 . Design Note (100–150 Words)

The sequence begins when the Customer inserts a card and enters a PIN. The ATM sends the card information and PIN to the BankSystem for validation. Withdrawal processing is allowed only when the PIN is valid. The ATM then forwards a request for PKR 5,000, and the BankSystem asks the Account to check the available balance. The alt fragment separates sufficient-funds and insufficient-funds outcomes. When funds are sufficient, the BankSystem requests a debit and waits for confirmation before sending approval to the ATM. Only then does the ATM dispense cash and display success. If funds are insufficient, the ATM displays an appropriate message and neither debit nor cash dispensing occurs. If the PIN is invalid, the ATM displays an error and stops the withdrawal flow. The ATM is assumed to have cash and network service.